

LIQUID LENS

Aim:

To find the refractive index of a liquid using convex lens and plane mirror

Apparatus:

A plane mirror, a convex lens, an optical needle, a clamp stand, a dropper and a half metre scale.

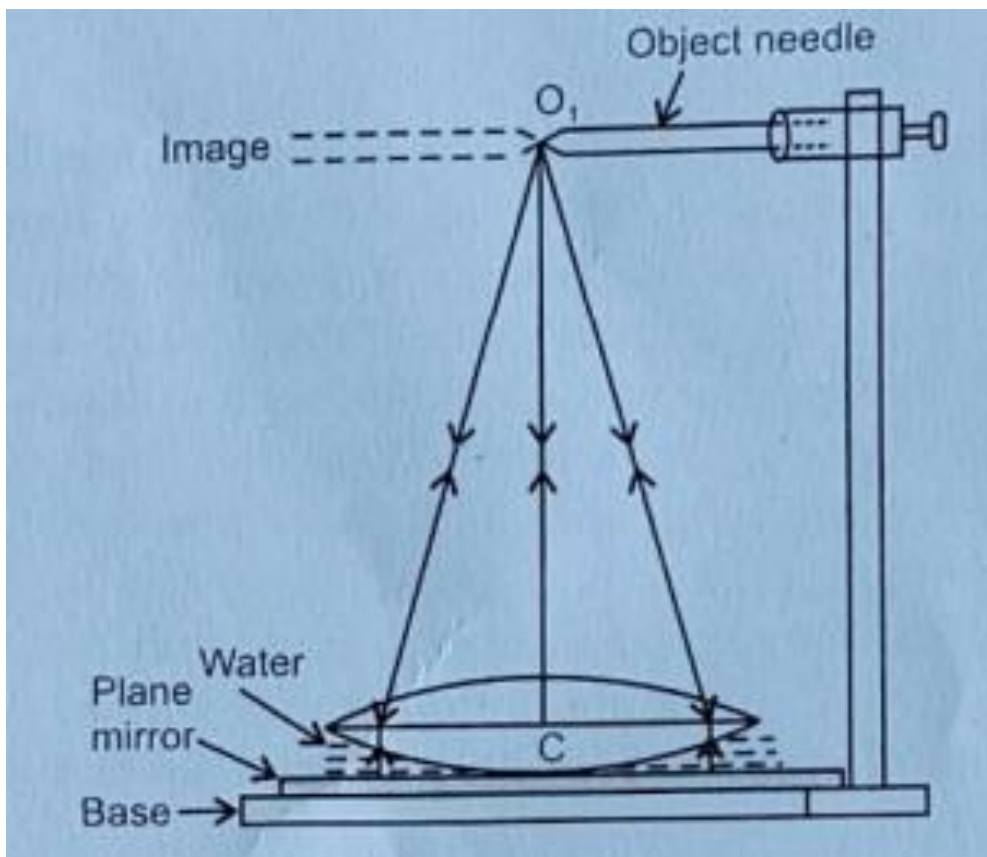
Theory:

Refractive index of liquid, $n = 2 - \frac{f_g}{F}$

Where, f_g - focal length of convex lens

F- focal length of compound lens formed by convex lens of glass and plano-concave lens of water

Diagram:



Procedure:

1. Place the plane mirror upon the horizontal and flat base of vertical stand. Place a convex lens on the reflecting surface of the plane mirror.

2. Clamp on object needle in the vertical rod of the iron stand so that its tip is vertically above the optical centre of the convex lens (L).

3. Keeping the eye at least at a distance of distinct vision adjust the position of the object needle till parallax between object needle and its inverted image is removed tip to tip. Note the distance 'a' between object needle and the upper surface of the convex lens and 'b', the distance between the object needle and the upper surface of plane mirror using plumb line and half metre scale.

4. Now with the help of dropper put a few drops of experimental liquid, i.e., water on the plane mirror under the lens. See that no air bubbles are introduced in water in this process. Again adjust the position of the object needle by raising it in upward direction. Hence, find the new position of the object needle where the parallax between object needle and its inverted image is removed tip to tip. Measure the distance between the object needle and the upper surface of the convex lens, and between the object needle and the surface of the plane mirror using a plumb line and a metre scale.

5. Repeat the procedure at least two more times and record the observations in the table given below.

Observations:

Rough focal length of given convex lens =

TABLE FOR FOCAL LENGTH OF CONVEX LENS, f_g AND COMPOUND LENS, F

S. No.	Convex lens placed on plane mirror		Focal length of convex lens $f_g = \frac{a+b}{2}$ (cm)	Convex lens placed on plane mirror with water in between		Focal length of combination $F = \frac{a_1+b_1}{2}$ (cm)
	Distance between object needle and			Distance between object needle and		
	Lens a (cm)	Plane mirror b (cm)		Lens a ₁ (cm)	Plane mirror b ₁ (cm)	
1						
2						
3						
Mean fg =				Mean F =		

Calculations:

Refractive index of water, $n = 2 - \frac{f_g}{F}$

=

Result:

The refractive index of water using a convex lens and a plane mirror =

Precautions:

1. Before starting the experiment, clean the convex lens and the plane mirror.
2. Clamp the object needle horizontally so that its tip touches the principal axis normally.

Sources of error:

1. The base of the plane mirror may not be horizontal.
2. The object needle may not be horizontal.